Chenglong Wang (王成龙)

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EDUCATION

University of Chinese Academy of Sciences 2019–2022

M.S. in Computer Science | Exam-Free Postgraduate Recommendation

Thesis: Adaptive Doudizhu Algorithm Based on Deep Monte Carlo and simulated annealing

Xi'an Jiao Tong University, Qian Xuesen Honors College 2015–2019

B.S. in Physics & minor in Mathematics from Young Gifted Program

GPA: 88.62/100 (First-Class Honours)

University of Science and Technology of China(USTC) 2017–2018

Visiting Studentship (C9 League Exchange Program)

SELECTED HONORS & AWARDS

- Second prize of Chinese Physics Olympiad, 2015
- First prize of China Undergraduate Physics Tournament (CUPT), 2016
- First prize of Mathematics competition of Chinese College Student, 2016
- First prize of China Undergraduate Mathematical Contest in Modeling, 2017
- National Scholarship of XJTU, 2019
- The champion of China Computer Game Championship in Doudizhu and Mahjong, 2022
- Outstanding Master's Thesis, 2022

Work Experience & Internships

Ubiquant investment Co., Ltd 2022–present(full time)

Quantitative algorithm Engineer

Multi-Factor Alpha & GAT-Based Stock Representation

Cross-Disciplinary Insight: Applied convex optimization and Lagrange multipliers (mathematics) for constrained portfolio construction under transaction cost penalties and regulatory position limits.

- Identified a structural blind spot in traditional statistical-arbitrage models, stocks treated as isolated one-hot entities with no inter-asset topology and addressed it by embedding 4,000+ A-share securities into dense low-dimensional representations via a Graph Attention Network (GAT) with multi-head attention, capturing sector co-movement, supply-chain links, and style-factor clusters simultaneously.
- Integrated GAT-derived embeddings with 200+ engineered multi-factor features (momentum, value, quality) as inputs to LGBM / XGBoost cross-sectional return-forecast models, enabling dynamic long-short portfolio construction with integrated risk budgeting for Kaggle Ubiquant Market Prediction (8/2893).

Asian Option Pricing & Derivatives Risk

Cross-Disciplinary Insight: Applied counterfactual regret minimization (competitive game theory) to balance exploration-exploitation in non-stationary market environments with changing volatility regimes.

- Developed a Monte Carlo simulation engine for Asian option pricing, incorporating antithetic variates and control variates to reduce variance; cut pricing error by 22% relative to the Black-Scholes benchmark on path-dependent payoffs.
- Deployed the model into the team's options desk after internal validation; portfolios guided by the new pricing framework outperformed the benchmark by 12% in cumulative returns over the subsequent deployment period.

Deep RL for Intraday Futures Execution

Technical Innovation: Bridged offline RL (Transformer architecture from NLP) with financial market microstructure, treating execution as a conditional sequence generation problem.

- Framed intraday futures trading as a sequential decision problem under market uncertainty, drawing on expertise from Master's thesis research on Nash equilibrium in imperfect-information games.
- Deployed PPO and Soft Actor-Critic agents to optimize dynamic position sizing across 10+ commodity futures contracts, incorporating market-impact models and transaction-cost penalties; achieved 15% improvement in risk-adjusted PnL and 22% reduction in execution slippage versus baseline heuristics.
- Implemented a offline Decision Transformer trained on 2+ years of historical order-flow data to learn optimal execution policies offline, eliminating live trial-and-error; reduced average transaction costs by 18% through learned order-splitting and intraday timing strategies.

Alibaba Co., Ltd 2021.05–2021.09

Intern in Taobao BU for automated dialogue systems

- Designed a knowledge-base-augmented response framework unifying classical IR (TF-IDF, BM25) with neural semantic matching (BERT embeddings), applying mathematical optimization to balance precision-recall tradeoffs at scale.
- Achieved 23% accuracy improvement over baseline rule-based systems; deployed into Taobao customer service handling 5M+ daily queries.
- Granted Chinese Invention Patent CN 110019715 B — Method for Response Determination in Knowledge-Augmented Dialogue Systems (issued Jul. 14, 2023).

SenseTime Research 2019.07–2020.02

Intern in Computer Vision Research working with Dr. Yu Liu :

- Proposed Incremental Hint Learning, a knowledge-distillation method that decouples feature-hint learning from detection supervision via gradual learning schedules, resolving out-of-distribution degradation in compressed face recognition models.
- Achievement: Team built world's #1 face recognition system (ICCV 2021 MFR Champion, NIST FRVT Leaderboard #1), processing billions of identity verifications globally.

RESEARCH EXPERIENCE

Institute of Automation, Chinese Academy of Sciences 2020–2022

Advised by Prof. Junliang Xing in imperfect information games for thesis:

- Integrated simulated annealing with counterfactual regret minimization (CFR), introducing temperature scheduling to improve the theoretical time complexity of regret value convergence speed for approximate Nash equilibrium of double game.
- Combined Deep Monte Carlo methods and simulated annealing for Doudizhu, incorporating residual networks to enhance stability, using stochastic prioritized sampling to improve sample efficiency, and applying multi-agent learning to strengthen the Actors performance.
- **Impact:** Achieved 54.4% win rate vs. SOTA DouZero algorithm, Won champion in China Computer Game Championship (Doudizhu & Mahjong, 2022).

Key Laboratory for Quantum Optics and Fine Mechanics of CAS, USTC 2017–2018

Advised by Prof.Wenlin Gong in Computational Physics:

- Helped develop theoretical models and performed simulations to evaluate how light source fluctuations affect computational ghost imaging (CGI), proposing real-time correction methods that improved image quality significantly under unstable lighting conditions.
- Created and optimized the GISCNL algorithm for single-detector color ghost imaging, integrating sparsity constraints and non-local self-similarity, which led to superior image quality with fewer measurements.
- Developed a super-resolution imaging technique that surpasses the diffraction limit using sparsity-constrained optimization and sparse speckle illumination, enabling stable reconstructions even at low signal-to-noise ratios (SNR < 10 dB).

__C. Wang__, W. Gong, et al. ," Influence of Receiving Numerical Aperture and Rough Target Size on Ghost Imaging via Sparsity Constraint. Chinese Laser," 2019, 46(08):270-275. (in Chinese)

P. Wang, L. Wei, __C. Wang__, et al., "Super-resolution imaging via sparsity constraint and sparse speckle illumination," Chinese Physics B 27(7), 074202 (2018)

P. Wang, __C. Wang__, et al., "Color ghost imaging via sparsity constraint and non-local self-similarity," Chinese Optics Letters 19, 021102 (2021)

X. Mei, __C. Wang__, et al., "Influence of the source's energy fluctuation on computational ghost imaging and effective correction approaches," Chinese Optics Letters 18, 042602 (2020)

State Key Laboratory of Multiphase Flow in Power Engineering, XJTU 2016–2018

Advised by Prof.Guanghui Su in Computational Physics:

- Coupled finite element method (FEM) and finite difference method (FDM) simulation of high-temperature heat pipe startup dynamics with Cotter's limit theory validation demonstrates successful thermal radiator performance for space reactor applications.

__Wang Chenglong__, Hu Lingwen, Qiu Suizheng, Su Guanghui, et al. , "Uncertainty Analysis of Transportable Fluoride-Salt-Cooled High Temperature Reactor (TFHR)," Nuclear Power Engineering 38(3), 168-171 (2017) *(in Chinese)*

Zhang Wenwen, __Wang Chenglong__, et al., "Thermal Safety Analysis of New Type Heat Pipe Reactor Core," Atomic Energy Science and Technology 51(5), 822-827 (2017) (in Chinese)

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Research Statement

The question that has driven my research across every domain I have worked in is the same: what is the right mathematical model for this problem? Not whether a model works empirically, but whether it correctly identifies the structural properties of the phenomenon being studied—the conservation laws, the information constraints, the decision-theoretic geometry—and translates them into a computational framework precise enough to make reliable predictions. My path from computational physics through game AI to quantitative finance has been a series of attempts to answer that question for increasingly complex and poorly-specified real-world systems.

Computational Physics and Modeling

My earliest research confronted this question in settings where the physical model was known but computationally demanding. As part of a joint XJTU–MIT project on the Transportable Fluoride-Salt-Cooled High Temperature Reactor (TFHR), I worked on coupling finite element and finite difference methods to simulate high-temperature heat pipe startup dynamics for space reactor applications. The modeling challenge was to translate the physical conservation laws governing heat transfer—energy balance at phase boundaries, fluid dynamics in the vapor core, conduction through the wall—into a discretized system that remained numerically consistent and stable under the extreme thermal gradients of startup transients. The coupling between the FEM and FDM domains had to respect the physical constraints at their interface precisely.

The second thread of my undergraduate research, at the Key Laboratory for Quantum Optics at USTC, posed a structurally different modeling problem: recovering images from indirect, incomplete measurements. Ghost imaging via sparsity constraints is fundamentally an inverse problem—the forward model (the linear sensing operator determined by the illumination patterns) is known, and the task is to invert it given far fewer measurements than unknowns. What makes this tractable is that natural images are sparse in appropriate transform domains; the correct model must encode this prior explicitly. I worked on the GISCNL algorithm, which extended the standard sparsity framework by incorporating non-local self-similarity as an additional structural constraint, enabling accurate single-detector color image reconstruction with substantially fewer measurements than competing methods. Encoding this prior correctly in the optimization objective is what made the approach work.

Modeling Learned Representations: Knowledge Distillation at SenseTime

During my internship at SenseTime Research, I encountered a modeling problem in deep learning that had the same flavor: standard knowledge distillation fails to transfer the structural properties of

a large teacher network to a compact student, because the standard formulation models only the teacher's output distribution, not the representational hierarchy that produces it. The practical consequence was that compressed face recognition models degraded on out-of-distribution inputs—unusual lighting, extreme pose, aging—precisely the inputs that depend on intermediate representational structure rather than surface statistics.

The solution I developed, Incremental Hint Learning, required first building a precise model of what 'representational structure' means in a convolutional recognition network: a hierarchy of feature abstractions, each level encoding increasingly task-specific information, with the lower levels providing the geometric invariances that enable generalization. The distillation curriculum was then designed to transfer this structure level by level—a direct consequence of modeling the dependency structure of the learned representation correctly, not a heuristic. The approach contributed to the team achieving first place in the ICCV 2021 Masked Face Recognition Challenge and the NIST Face Recognition Vendor Testing leaderboard.

Modeling Strategic Interaction: Doudizhu and Imperfect-Information Games

My master's thesis at the Chinese Academy of Sciences required building a precise model of a different class of real-world problem: multi-agent decision-making under hidden information. Doudizhu is a three-player card game with asymmetric roles, stochastic card distributions, and partially observable state—a setting where the standard RL formulation of a single agent optimizing a known reward function is insufficient. The correct model had to account for the cooperative structure of the landlord-versus-peasants asymmetry, the strategic interaction between agents whose interests partially conflict, and the hidden information that forces inference about opponents' hands.

The algorithmic framework I developed combined deep Monte Carlo tree search with a regret-value-matching procedure adapted with simulated annealing temperature scheduling to improve convergence toward approximate Nash equilibria. Each design choice reflected a modeling decision: the Monte Carlo rollouts encode the stochastic structure of hidden-card draws; the regret-matching procedure models the game as an extensive-form game and targets the correct game-theoretic solution concept; the simulated annealing schedule addresses the non-convexity of the convergence landscape. The system achieved a 54.4% win rate against the prior state-of-the-art and won the 2022 China Computer Game Championship in both Doudizhu and Mahjong.

Modeling Financial Systems: Quantitative Research at Ubiquant

Three years at Ubiquant Investment have posed the modeling challenge in its hardest form: financial markets are high-dimensional, non-stationary, and only partially observable, and the cost of a wrong model is immediate and quantifiable. Each project has required identifying the mathematical structure of the problem before choosing a computational approach. For cross-sectional equity prediction, the structural insight was that stocks are not isolated entities but nodes

in a heterogeneous relational graph—connected by sector membership, supply-chain links, and shared style-factor exposures. Encoding this topology via a Graph Attention Network, rather than treating securities as exchangeable feature vectors, reflected a precise modeling commitment about the information structure of the return-generating process; the system ranked 8th out of 2,893 teams on Kaggle's Ubiquant Market Prediction benchmark.

For Asian option pricing, the modeling question was how to handle path dependency correctly. Black-Scholes closed-form approximations are exact only for European payoffs; path-dependent instruments require Monte Carlo simulation over the full price path. I built a simulation engine incorporating antithetic variates and control variates—variance reduction techniques that exploit the geometric structure of the payoff function to concentrate samples where they matter most—reducing pricing error by 22% against the benchmark. For execution optimization, framing intraday futures trading as a sequential decision problem made RL the natural framework; when online exploration became prohibitive, treating experienced traders' execution logs as expert demonstrations and training a Decision Transformer via offline imitation learning was a direct consequence of modeling the problem structure correctly. Transaction costs fell 18%.

Research Direction

The pattern across these projects is consistent: the quality of the solution is bounded by the precision of the model. Where I have succeeded, it has been because the mathematical framework correctly identified the structural properties of the real phenomenon. Where I have encountered limits, it has been because the modeling framework was incomplete: the execution agents lack a principled account of how market microstructure constrains feasible actions; the alpha model has no formal mechanism for detecting regime change. These are the modeling problems I want to pursue in a PhD.

Chenglong Wang

March 2026

西安交通大學
XI'AN JIAOTONG UNIVERSITY

本科生成绩单
Transcript of Undergraduate Student

姓名: 王成龙
Name: Wang Chenglong
学院: 理学院
School: School of Science

入学年月：2015年08月
Date of Enrollment: August 2015

学号：2150900081
Student No.: 2150900081

出生日期：1998年01月01日
Date of Birth: January 01,1998

专业：理科试验班（物理H）
Major: Honors science Program (Physics)

毕业年月：2019年07月
Date of Graduation: July 2019

学制：4年制
Length of Schooling: Four Years

性别：男
Gender: Male

专业班：物理试验班51

绩点：3.71 学分成绩：88.62
GPA: 3.71 Average Score: 88.62

课程 Course	学分 Credit	成绩 Score	绩点 GPA
第一学年（2015-2016）第一学期 1st Academic Year（Semester 1）（2015-2016）			
高等数学II1 Advanced Mathematics II1	6	95	4.3
经典力学 classical mechanics	3	90	4.0
线性代数与解析几何II Linear Algebra and Analytic Geometry II	3.5	97	4.3
普通物理实验1 physics experiment of university 1	1	85	3.7
思想道德修养与法律基础 Moral and Legal Education	3	82	3.3
大学计算机基础II Fundamentals of Computers II	3	79	3.0
大学英语III1 College English III1	2	72	2.3
军训 Military Training	1	75	2.7
体育1 Sports 1	0.5	92	4.0
专业课程导论 introduction to major courses	1	通过 Pass	
第一学年（2015-2016）第二学期 1st Academic Year（Semester 2）（2015-2016）			
高等数学II2 Advanced Mathematics II2	6	96	4.3
热学 Thermal physics	3	99	4.3
毛泽东思想和中国特色社会主义理论体系概论1 Mao Zedong Thought and Introduction to the theoretical system of socialism with Chinese characteristics1	2	80	3.0
普通物理实验2 physics experiment of university 2	1	83	3.3
大学英语四级(CET4) College English Test(CET4)		450	
大学英语III2 College English III2	2	77	2.7
大学化学I College Chemistry I	4	86	3.7
国防教育 National Defence Education	1	71	2.0
体育2 Sports 2	0.5	97	4.3
中国近现代史纲要 Outline of Modern Chinese History	2	76	2.7
第二学年（2016-2017）第一学期 2nd Academic Year（Semester 1）（2016-2017）			
电路 Circuits	4.5	93	4.0
电磁学 electromagnetism	4	81	3.3
毛泽东思想和中国特色社会主义理论体系概论2 Mao Zedong Thought and Introduction to the theoretical system of socialism with Chinese characteristics2	2	86	3.7
概率论与数理统计 Probability Theory and Mathematical Statistics	3	91	4.0
普通物理实验3 physics experiment of university 3	1	83	3.3
数学物理方法 method of mathematical physics	4	100	4.3
微观经济学 Microeconomics	2	94	4.0
学术英语听说 Academic Listening and Speaking	2	70	2.0
光学 Optics	4	98	4.3
体育3 Sports 3	0.5	91	4.0
24式太极拳 24-Style Tai Ji Quan		60	
第二学年（2016-2017）第三学期 2nd Academic Year（Semester 3）（2016-2017）			
社会学概论 Introduction to Sociology	2	83	3.3
电工实习 Electrician Practice	1	A -	3.7
文物鉴赏 Appreciation of Cultural Relics	2	76	2.7
第三学年（2017-2018）第一学期 3rd Academic Year（Semester 1）（2017-2018）			
量子力学 Quantum Mechanics	5	98	4.3
近代物理实验1 Modern physics experiment1	1	88	3.7
计算物理 Computational Physics	3	84	3.3
热力学与统计物理 Thermodynamics and Statistical Physics	5	95	4.3
材料科学基础 I Foundation of Materials Science I	3	99	4.3
半导体材料与器件 Semiconductor Materials and Devices	2	92	4.0
光电子学 Opto-Electronics	4	93	4.0
第三学年（2017-2018）第二学期 3rd Academic Year（Semester 2）（2017-2018）			
近代物理实验2 Modern physics experiment2	1	87	3.7
粒子物理导论 Introduction to Elementary Particles	2	76	2.7
固体物理 Solid State Physics	4	75	2.7
第三学年（2017-2018）第三学期 3rd Academic Year（Semester 3）（2017-2018）			
专业实习 Specialized Practice	1	A -	3.7
第四学年（2018-2019）第一学期 4th Academic Year（Semester 1）（2018-2019）			
量子场论导论 Quantum Field Theory	3	89	3.7
量子光学与量子信息导论 Introduction of quantum optics and quantum information	2	91	4.0
英语写作 English Writing	2	77	2.7
群论导论 Introduction to Group Theory	2	78	3.0
C++程序设计 C++Programming	3	91	4.0
第四学年（2018-2019）第二学期 4th Academic Year（Semester 2）（2018-2019）			
课外实践8学分 8 credit	8	通过 Pass	
综合性实践训练 Comprehensive practice training	1	90	4.0
毕业设计	10	A	4.0

百分制 Centesimal Grade	95~100	90~94	85~89	81~84	78~80
等级 Grades	优+(A+)	优(A)	优-(A-)	良+(B+)	良(B)
绩点 GPA	4.3	4.0	3.7	3.3	3.0
75~77	72~74	68~71	64~67	60~63	0~59
良-(B-)	中+(C+)	中(C)	中-(C-)	及格(D)	不及格(F)

标签页 1

references

Name	Email	Laction	Relation
Prof. Junliang Xing	lxing123@iacas.dpdns.org	Institute of Automation, Chinese Academy of Sciences	Master Thesis advisor
Dr. Byran Dai	byrandai@ubiquant.eu.cc	Ubiquant investment Co., Ltd	Leader researcher
Prof. Guoping Guo	jackiewang@ustc.edu	University of Science and Technology of China(USTC)	Undergraduate Thesis advior